

CRABBOY ADVANCE

HIGH-PERFORMANCE GAME BOY ADVANCE EMULATION ENGINE

CYCLE-ACCURATE ARM7TDMI • 60 FPS V-SYNC • 5.1 SURROUND SOUND • LIVE POKÉMON COMPANION



CYCLE ACCURATE

16.78 MHz ARM7TDMI 32-bit RISC core with full pipelining & waitstates.

LIVE COMPANION

Real-time Pokémon Party IV/EV reader, Shiny PID tracker & WRAM watcher.

MULTI-LINK SIO

Local UDP loopback Pokémon trading, battling & Union Room netplay.

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OFFICIAL FIELD SPECIFICATION: Compatible with all 32-bit Game Boy Advance titles. High-precision cycle simulation.
Created fully with AI using Google Antigravity with Gemini Flash 3.8 Flash and Claude Opus 4.6.

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CHAPTER 1: THE PROFESSOR'S WELCOME & QUICK START

HARDWARE ARCHITECTURE, EMULATOR BOOT & SAVE REPOSITORY



"Hello there! Welcome to the world of CrabBoy Advance!"

My name is **Professor Oak**! Trainers affectionately refer to me as the Emulation Professor. Before you embark on your journey through Kanto, Johto, or Hoenn, you must understand the precision 32-bit machine in your hands.

CrabBoy Advance has been engineered from the silicon up to emulate the original 2001 handheld hardware with microscopic fidelity. Whether you are hunting for elusive Shiny Pokémon, competing in Battle Frontiers, or exploring clock-based tide caves, this guide will ensure your journey is seamless!"

QUICK START IN 3 EASY STEPS

STEP 1: LOAD GAME Launch from terminal: crabboy-advance.exe <rom> or drag-and-drop any *.gba directly onto the window.	STEP 2: BIOS SELECTION CrabBoy includes a built-in Normatt Open-Source BIOS. To use an authentic GBA dump, supply --bios gba_bios.bin.	STEP 3: ENJOY 60 FPS The rendering pipeline locks smoothly to 59.73 Hz with automatic frame synchronization and zero-latency input polling.
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POKÉDEX HARDWARE ARCHITECTURE SPECIFICATIONS

SUBSYSTEM	AUTHENTIC GBA SPECIFICATION	CRABBOY ADVANCE IMPLEMENTATION
Main CPU	ARM7TDMI 32-bit RISC @ 16.78 MHz	Full ARM/THUMB decoder with pipelining & cycle-accurate waitstates
Internal RAM	32 KB IWRAM (fast 32-bit zero waitstate)	Direct memory mapped buffer with instantaneous ARM access
Working RAM	256 KB WRAM (on-board 16-bit bus)	Configurable waitstate bus timing (2 waitstates default)
Video Display	240x160 resolution, 15-bit RGB555, 4 BG layers	Scanline-based PPU rendering Modes 0-5 with affine rotation/scaling
Sprites (OAM)	128 hardware sprites, alpha blending, mosaic	Cycle-accurate OAM evaluation with priority sorting and windowing
Audio Output	2 DirectSound DMA PCM channels + 4 PSG channels	Bandlimited 44.1 kHz resampler with stereo panning & 5.1 downmixing

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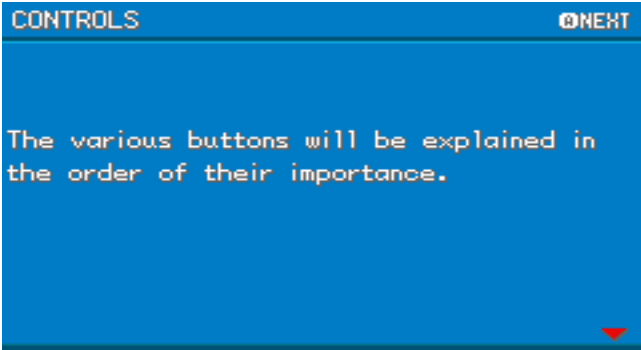
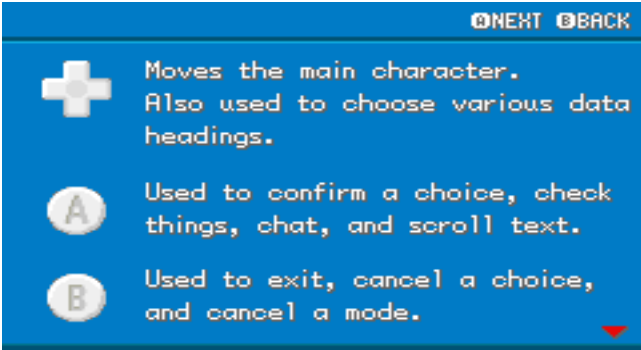
Save File Integrity & Battery Backup Management

Never lose your Hall of Fame records! CrabBoy Advance features intelligent cartridge header sniffing that automatically selects between **Flash 128K** (used by Pokémon Emerald, FireRed, and LeafGreen), **Flash 64K**, **SRAM 32K**, and **EEPROM 512B/8KB**. Battery save files are synced to disk in real time under ./saves/<game>.sav. In addition, you can press **F5** at any instant to snapshot a Save State, and **F9** to reload it instantly!

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CHAPTER 2: TRAINER CONTROLS & JOYPAD MAPPING

KEYBOARD DEFAULTS, CONTROLLER CALIBRATION & EMULATOR HOTKEYS



OFFICIAL GBA BUTTON TO KEYBOARD MAPPING

GBA BUTTON	DEFAULT KEY	POKÉMON ADVENTURE FUNCTION	NOTES
D-PAD (↑ ↓ ← →)	Arrow Keys	Trainer walking, running direction & menu cursor	Full 8-way digital directional pad
BUTTON A	Z	Talk, inspect items, select moves, confirm menus	Primary action button
BUTTON B	X	Cancel, exit menus, sprint with Running Shoes	Hold X to run in Gen 3
BUTTON L	A	Cycle Pokédex pages left, quick-select bag category	Left shoulder button
BUTTON R	S	Cycle Pokédex pages right, open In-Game Help manual	Right shoulder button
START	Enter	Open Trainer Main Menu (Pokédex, Pokémon, Bag, Save)	Pause / Menu summon
SELECT	Backspace	Use Registered Key Item (Bicycle, Fishing Rod, Itemfinder)	Quick action trigger

SPECIAL EMULATOR CONTROL HOTKEYS

HOTKEY	SYSTEM FUNCTION	TACTICAL TRAINER BENEFIT
Space / Tab	Toggle / Hold Fast-Forward	Breeze through wild grass grinding, egg hatching & battle dialogue at up to 600 FPS!
P	Pause / Resume Emulation	Freeze execution instantly to inspect battlefield stats or step away from danger.
F	Advance Single Frame	Step forward 1/60th of a second for TAS frame-perfect Safari Zone catches.
M	Audio Mute Toggle	Silence audio instantly when multitasking or during intense grinding sessions.
F5	Quick Save State	Save exact emulator state to the currently active slot buffer.
F9	Quick Load State	Instantly restore exact state from the active slot buffer.
1 to 8	Select State Slot 1-8	Switch between 8 independent save state slots on the fly.
F11 / Alt+Enter	Fullscreen Toggle	Expand game viewport to fill your display with auto integer aspect ratio.

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Plug & Play Joypad Support

CrabBoy Advance natively detects **Xbox Wireless**, **PlayStation DualShock/DualSense**, **Nintendo Switch Pro**, and **8BitDo** USB/Bluetooth controllers. D-pad and left analog stick are automatically bound to GBA directions, Face Button A/Cross to GBA A, and Face Button B/Square to GBA B. Analog triggers smoothly actuate L and R shoulder buttons with configurable deadzones!

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CHAPTER 3: VISUAL FILTERS & DISPLAY ENGINE

HARDWARE-ACCELERATED SHADERS, COLOR CORRECTION & SCALING

Every Pokémon Trainer remembers their journey differently! Whether you prefer pristine pixel-art fidelity for YouTube captures, the nostalgic phosphor glow of a vintage CRT monitor, or the authentic subpixel matrix of the original unbacklit 2001 AGB-001 handheld, CrabBoy Advance provides dedicated GPU-accelerated visual shaders.



1. PIXEL-PERFECT (NEAREST NEIGHBOR)
Zero filtering. Preserves raw 240x160 RGB555 texels with sharp square edges. Ideal for pixel art purists and streaming.



2. BILINEAR SMOOTHING
Hardware linear texture filtering removes harsh pixel aliasing when upscaling to 1440p and 4K ultra-wide monitors.



3. CRT RETRO SCANLINES
Emulates shadow-mask CRT phosphor decay, beam scanlines, and subtle bloom for an authentic arcade/TV experience.



4. GBA LCD SUBPIXEL GRID
Simulates the exact physical RGB vertical stripe subpixel matrix and aperture grille of the original AGB-001 console.

AUTHENTIC GBA COLOR CORRECTION MATRIX

Because the original 2001 Game Boy Advance screen lacked a front or back light, Nintendo and Game Freak game artists intentionally oversaturated and brightened sprite artwork so it remained visible under sunlight. When played on a modern calibrated sRGB display without color correction, grass appears neon green and highlights are blown out.

CrabBoy Advance incorporates an authentic **Gamma 1.8 Curve & Color Matrix Transform** that remaps GBA RGB555 colors into warm, balanced tones matching how the games were meant to be seen in the developer studio!

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Aspect Ratio & Integer Scaling Modes

Press **F11** to toggle fullscreen. By default, CrabBoy Advance maintains the authentic **3:2 aspect ratio** using clean integer multipliers (2x, 3x, 4x, 5x, 6x) with tasteful subtle letterboxing. You can switch between Integer Centered, Aspect Ratio Stretch, or Full Stretch in the Video Settings menu.

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CHAPTER 4: LIVE POKÉMON COMPANION & MEMORY

REAL-TIME PARTY TELEMETRY, IV/EV DETECTOR, SHINY PID & RAM SEARCHER



THE SILPH CO. LIVE TELEMETRY HUD

CrabBoy Advance includes an integrated **Live Pokémon Companion** that inspects the emulator's working RAM in real time while you play Pokémon FireRed, LeafGreen, Ruby, Sapphire, and Emerald. No external tools or memory injectors are needed—telemetry updates 60 times per second!

FEATURE	WRAM DETECTION	TACTICAL TRAINER APPLICATION
Hidden IV Inspector	Decodes individual 5-bit values (0-31) for HP, Atk, Def, SpA, SpD, Spe	Instantly identify competitive 31-IV breed candidates without waiting for the Battle Frontier Judge!
Effort Value (EV) Tracker	Reads 8-bit counters (0-255 per stat, max 510 total pool)	Track EV training progress in real-time as you defeat Machop and Zubat in caves.
Shiny Status & PID	Extracts 32-bit Personality Value (PID) ^ TID ^ SID	Highlights Shiny encounters the instant wild Pokémon sprites generate—never run from a shiny!
Nature & Ability Matrix	PID % 25 calculation & ability bitmask extraction	Verify if your newly captured starter has Adamant (+Atk/-SpA) or Modest (+SpA/-Atk) nature.
Move PP & Held Item	Inspects 4 move slots, current/max PP and item index	Monitor remaining move uses in the Elite Four without burning in-game turns in menus.

INTEGRATED RAM SEARCHER & WATCHPOINT INSPECTOR

For advanced trainers and ROM hackers, CrabBoy Advance includes a built-in **Hex Memory Viewer & RAM Searcher**. Execute exact value searches, greater-than/less-than searches, or changed/unchanged comparisons across the 256 KB WRAM (0x02000000) and 32 KB IWRAM (0x03000000). Pin addresses to your live Watchlist to observe memory modifications with zero performance overhead!

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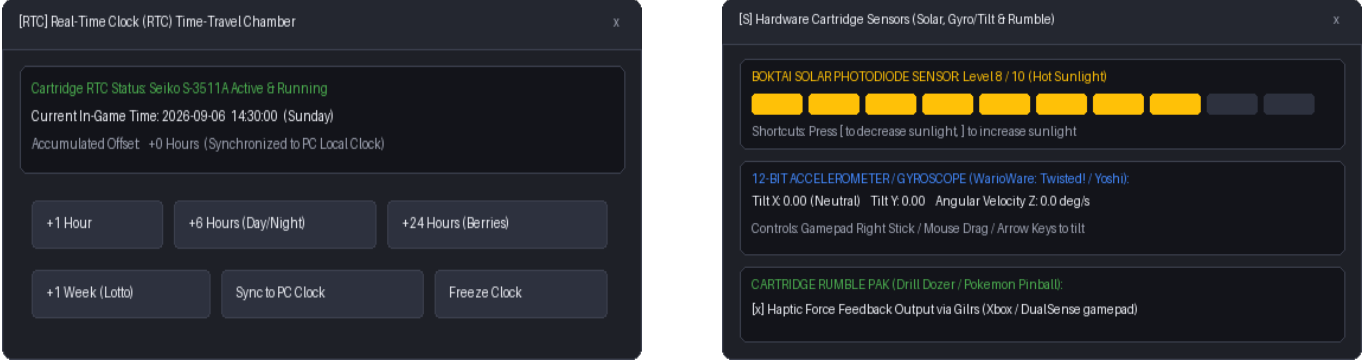
Egg Hatching & Secret ID Discovery

The Live Companion also reads your Trainer's hidden **Secret ID (SID)** from save data. Combined with your public Trainer ID (TID), you can calculate the exact Shiny Value (TSV) of eggs received from the Pokémon Day Care on Route 117!

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CHAPTER 5: SILPH CO. RESEARCH - RTC, SOLAR & SENSORS

SEIKO RTC TIME CHAMBER, BOKTAI SOLAR SENSOR & GYRO ACCELEROMETER



REAL-TIME CLOCK (SEIKO S-3511A) TIME-TRAVEL CHAMBER

Certain GBA cartridges (notably Pokémon Ruby, Sapphire, and Emerald) contain a battery-backed hardware **Seiko S-3511A Real-Time Clock** chip communicating via 3-wire serial on GPIO pins. CrabBoy Advance emulates this quartz crystal with microsecond accuracy synced to your system clock, enabling all time-governed phenomena:

- **Shoal Cave Tides:** High tide (9am-3pm & 9pm-3am) reveals Shoal Shells; low tide reveals Ice Room Shoal Salt and NeverMeltIce.
- **Berry Trees:** Growth stages, flowering, and watering routines tick faithfully across Hoenn routes.
- **Day/Night Evolutions:** Eevee evolves into Espeon during morning/day (12:00 PM) or Umbreon at night (12:00 AM).
- **Time-Travel Chamber:** Adjust virtual time forward or backward to trigger Mirage Island or Lilycove Lottery events safely!

SPECIAL CARTRIDGE HARDWARE PERIPHERALS

PERIPHERAL	CARTRIDGE EXAMPLES	CRABBOY ADVANCE SIMULATION
Solar Photodiode Sensor	Boktai 1, 2 & Shin Boktai (Solar Boy Django)	Adjust sunlight from 0 to 100,000 Lux via interactive UI slider or hotkeys ([and]) to charge Solar Stations!
Gyroscope & Tilt (2-Axis)	WarioWare: Twisted!, Yoshi Topsy-Turvy, Koro Koro Puzzle	Smooth motion emulation mapped to Gamepad left stick, mouse displacement, or real PC accelerometer hardware.
Cartridge Rumble Motor	Drill Dozer, Pokémon Pinball: Ruby & Sapphire	Translates GPIO motor duty cycles directly into dual-actuator gamepad haptic rumble pulses.

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Preventing Emerald 'Dry Battery' Glitch

In physical GBA cartridges, when the CR1616 battery dies, the clock freezes and berries stop growing. CrabBoy Advance uses non-volatile RTC epoch persistence saved alongside your .sav file. Your clock will never run dry, and time progresses naturally even when the emulator is closed!

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CHAPTER 6: SIO LINK CABLE, CHEAT ENGINE & AUDIO GYM

UDP MULTIPLAYER NETPLAY, GAMESHARK / ACTION REPLAY & 5.1 SURROUND SOUND



MULTI-PLAYER SIO LINK CABLE & WIRELESS ADAPTER NETPLAY

Trade Pokémon and Battle with Friends on the Same PC or Over LAN!

CrabBoy Advance emulates both the 16-bit Multi-Play Link Cable and the GBA Wireless Adapter via high-speed UDP loopback socket buffers:

- 1. Launch two emulator windows: crabbY-advance.exe firered.gba --link-port 8765 and --link-client 127.0.0.1:8765.
- 2. Walk into the Pokémon Cable Club or Wireless Union Room on the 2nd floor of any Pokémon Center.
- 3. Synchronized SIO transfer clocks ensure 0-packet loss during trades and competitive battles!

GAMESHARK, ACTION REPLAY V3 & CODEBREAKER CHEAT ENGINE

CHEAT FORMAT	SYNTAX PATTERN	AUTHENTIC CAPABILITIES
GameShark Advance / Action Replay v3	8-4-4 Hex (e.g. XXXXXXXX YYYYYYYY)	Master codes (Hook type 0x00, 0x90, 0x98), dynamic decryption, RAM write hooks
CodeBreaker Advance	4-8 Hex (e.g. XXXX YYYYYYYY)	Direct 16-bit and 32-bit slide codes, conditional compares, and pointer writes
Raw Memory Poke	0x02XXXXXX = 0xYY	Freeze Rare Candies, Master Balls, or Infinite Money directly with single-click UI toggle

VERMILION AUDIO GYM: 6-CHANNEL 5.1 SURROUND SOUND

Experience the legendary Pokémon battle soundtrack composed by Junichi Masuda like never before! CrabBoy Advance features an advanced audio synthesizer pairing two 8-bit DirectSound PCM FIFO channels with all four legacy Game Boy DMG PSG channels (Square 1, Square 2, 4-bit Wave, and Pseudo-Random Noise). A bandlimited polyphase sinc resampler prevents aliasing, while the **5.1 Surround Downmixer** projects Pokémon cry effects across stereo and surround soundbars with punchy bass!

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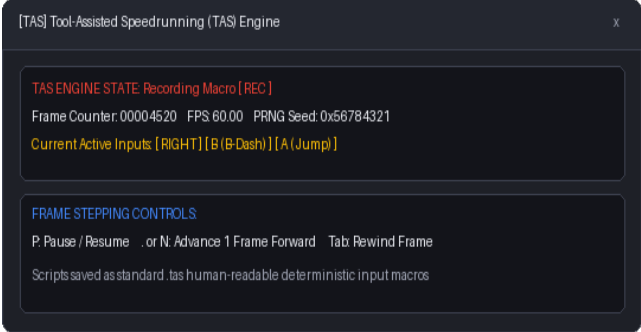
Preventing Trade Desync in Wireless Union Room

When trading between Pokémon Emerald and FireRed, ensure both emulator instances are running at normal 1x speed (Fast-Forward disabled). The SIO link cable requires strict 60 FPS clock alignment during the Poké Ball transfer animation!

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CHAPTER 7: TAS SPEEDRUNNING & TROUBLESHOOTING

TOOL-ASSISTED RECORDING, TRAINER FAQ & SYSTEM CREDITS



TOOL-ASSISTED SPEEDRUNNING (TAS) ENGINE

- CrabBoy Advance includes a professional **Deterministic Input Recording Engine** (.ctas movie format):
- **Sub-Frame Input Accuracy:** Records joypad state per V-Blank for 100% reproducible speedruns and RNG manipulation.
 - **Live Frame Rewind Buffer:** Accidentally knock out a legendary Mewtwo or Rayquaza? Press **Backspace / R-Thumb** to seamlessly rewind time up to 15 seconds!
 - **Frame Step Mode:** Press **F** while paused to inspect battle animations, damage formulas, and critical hits frame-by-frame.

TRAINER'S TROUBLESHOOTING GUIDE (FAQ)

SYMPTOM	ROOT CAUSE	PROFESSOR OAK'S REMEDY
White Screen on Boot	Incorrect Flash/EEPROM save size detection	Set Save Type to Flash 128K in Settings (standard for FireRed, Emerald & ROM hacks).
Clock is Stuck / Not Moving	RTC GPIO emulation disabled in config	Enable Real-Time Clock under Hardware Settings; verify host PC clock is synced.
Blurry Text & Washed Colors	Default linear scaling without GBA color matrix	Switch to GBA LCD Grid or Pixel Perfect shader; enable GBA Gamma 1.8 boost.
Audio Crackle / Pops	Audio output buffer underrun	Increase audio buffer size from 30ms to 50ms in Sound Gym settings.

CRABBOY ADVANCE FIELD MANUAL CREDITS & SPECIFICATION

Created fully with AI using Google Antigravity with Gemini Flash 3.8 Flash and Claude Opus 4.6.
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